

SANJIAZI, China

WMO: 507450

Lat: 47.3761N

Lon: 123.9236E

Elev: 148

StdP: 99.56

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.9	-25.8	-32.2	0.2	-26.3	-30.3	0.2	-24.6	7.9	-14.0	6.9	-14.4	1.2	250	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	32.2	20.9	30.4	20.5	29.0	20.2	24.0	28.0	23.0	27.2	22.2	26.7	3.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.8	17.8	26.4	21.7	16.7	25.5	20.7	15.6	24.8	73.4	28.1	69.5	27.1	66.2	26.8	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.3	6.3	-29.8	35.0	3.1	2.1	-32.0	36.5	-33.9	37.7	-35.6	38.9	-37.9	40.4		
			-29.9	25.4	3.1	1.1	-32.2	26.2	-34.0	26.9	-35.7	27.5	-37.9	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-17.7	-12.4	-3.4	7.1	15.2	21.4	23.8	21.8	15.5	6.1	-6.0	-15.5
	DBStd	15.28	4.23	5.13	5.89	4.82	4.24	3.63	2.64	2.71	3.93	4.65	6.07	4.53
	HDD10.0	3432	858	628	414	113	6	0	0	4	136	481	792	
	HDD18.3	5382	1116	862	673	338	116	12	1	4	102	378	731	1050
	CDD10.0	1517	0	0	1	26	168	343	428	366	168	17	0	0
	CDD18.3	424	0	0	0	1	20	104	171	112	17	0	0	0
	CDH23.3	3428	0	0	0	12	213	947	1395	749	110	1	0	0
	CDH26.7	1019	0	0	0	2	61	334	440	164	18	0	0	0
Wind	WSAvg	2.8	2.2	2.5	3.2	3.7	3.6	3.0	2.7	2.6	2.8	2.9	2.7	2.2
Precipitation	PrecAvg	440	2	2	6	17	31	71	138	91	45	18	4	4
	PrecMax	653	7	16	23	92	109	226	278	224	123	72	25	14
	PrecMin	250	0	0	0	0	0	7	10	5	4	0	0	0
	PrecStd	97	2	3	6	16	26	41	67	51	32	16	4	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.1	4.9	15.4	24.9	31.3	35.0	34.3	31.7	29.0	22.0	11.3	-0.8
		MCWB	-6.6	-0.4	6.1	11.8	16.3	20.3	22.2	21.6	19.1	12.9	4.8	-3.9
	2%	DB	-7.0	1.1	11.2	21.4	28.4	32.4	32.3	29.8	26.2	18.5	7.8	-4.2
		MCWB	-9.1	-3.1	3.9	9.9	15.2	19.7	22.3	21.8	17.7	11.0	2.7	-6.5
	5%	DB	-9.0	-1.7	8.3	18.7	25.6	30.1	30.5	28.5	24.3	15.7	4.7	-6.7
		MCWB	-10.7	-4.9	1.9	8.8	14.0	19.1	21.8	21.2	16.5	9.0	0.7	-8.4
	10%	DB	-11.0	-4.0	5.9	16.1	23.2	28.4	29.0	27.2	22.4	13.5	2.1	-8.7
		MCWB	-12.3	-6.6	0.4	7.4	12.8	18.5	21.4	20.6	15.4	7.8	-1.2	-10.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.8	-0.7	6.5	12.7	18.0	22.7	25.9	24.7	21.5	14.4	5.2	-3.4
		MCDB	-4.2	4.4	14.1	22.2	26.6	30.2	29.7	28.1	25.9	19.5	10.1	-1.1
	2%	WB	-8.9	-3.0	4.1	10.8	16.4	21.5	24.5	23.6	19.5	12.0	2.8	-6.4
		MCDB	-7.1	0.8	10.5	19.4	24.7	28.7	28.5	27.4	23.7	16.9	7.6	-4.4
	5%	WB	-10.7	-4.8	2.3	9.2	15.2	20.6	23.6	22.6	18.0	10.1	0.8	-8.4
		MCDB	-9.2	-1.8	7.9	17.5	23.2	26.9	27.7	26.7	22.4	15.0	4.3	-6.8
	10%	WB	-12.3	-6.6	0.6	7.8	14.0	19.9	22.7	21.7	16.6	8.2	-0.9	-10.2
		MCDB	-11.1	-4.1	5.4	15.3	21.9	25.8	26.9	25.8	21.0	12.8	1.7	-8.7
Mean Daily Temperature Range	5% DB	MDBR	10.0	11.1	10.9	11.5	10.9	9.5	8.1	8.4	10.1	9.8	8.6	8.9
		MCDBR	11.5	12.9	13.5	14.7	13.9	12.2	10.2	9.7	11.6	12.9	11.1	10.5
		MCWBR	9.8	9.8	8.1	7.5	5.8	4.2	3.6	3.8	5.3	7.0	7.2	8.6
	5% WB	MCDBR	11.2	12.6	12.9	13.3	11.6	9.7	7.7	7.9	9.0	11.3	10.7	10.4
		MCWBR	9.7	9.7	8.0	7.2	5.6	4.1	3.9	3.9	5.0	7.2	7.1	8.5
Clear-Sky Solar Irradiance	taub		0.289	0.311	0.373	0.465	0.532	0.564	0.529	0.473	0.413	0.384	0.325	0.299
	taud		2.216	2.181	2.075	1.912	1.811	1.843	1.976	2.103	2.181	2.133	2.211	2.176
	Ebn at Noon		777	840	832	784	745	724	745	772	784	735	718	704
	Edh at Noon		81	104	137	178	205	200	173	146	123	110	81	75
All-Sky Solar Radiation	RadAvg		1.95	3.11	4.37	5.22	5.80	5.79	5.51	5.11	4.31	3.05	1.92	1.47
	RadStd		0.17	0.22	0.29	0.34	0.43	0.60	0.59	0.38	0.36	0.16	0.13	0.14

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page